

Module 3

CHY1009 – Chemistry and Environmental Studies

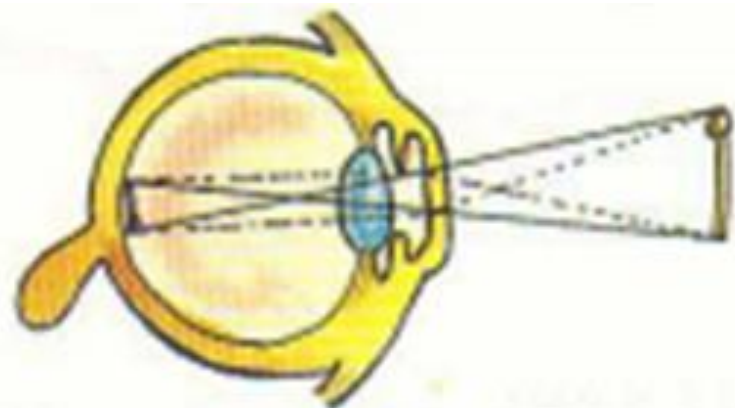
Electrochemistry and Chemical Sensors

Dr. Sabeel M Basheer
sabeel.basheer@vitap.ac.in

Our world is changing rapidly, and sensors play an important role in this process



Sensors in our body



We want to measure

Presence / Absence

Identity

Concentration

Qualitative or Quantitative

How can we measure?

Magnetic & Electric Fields

Resistance

Capacitance

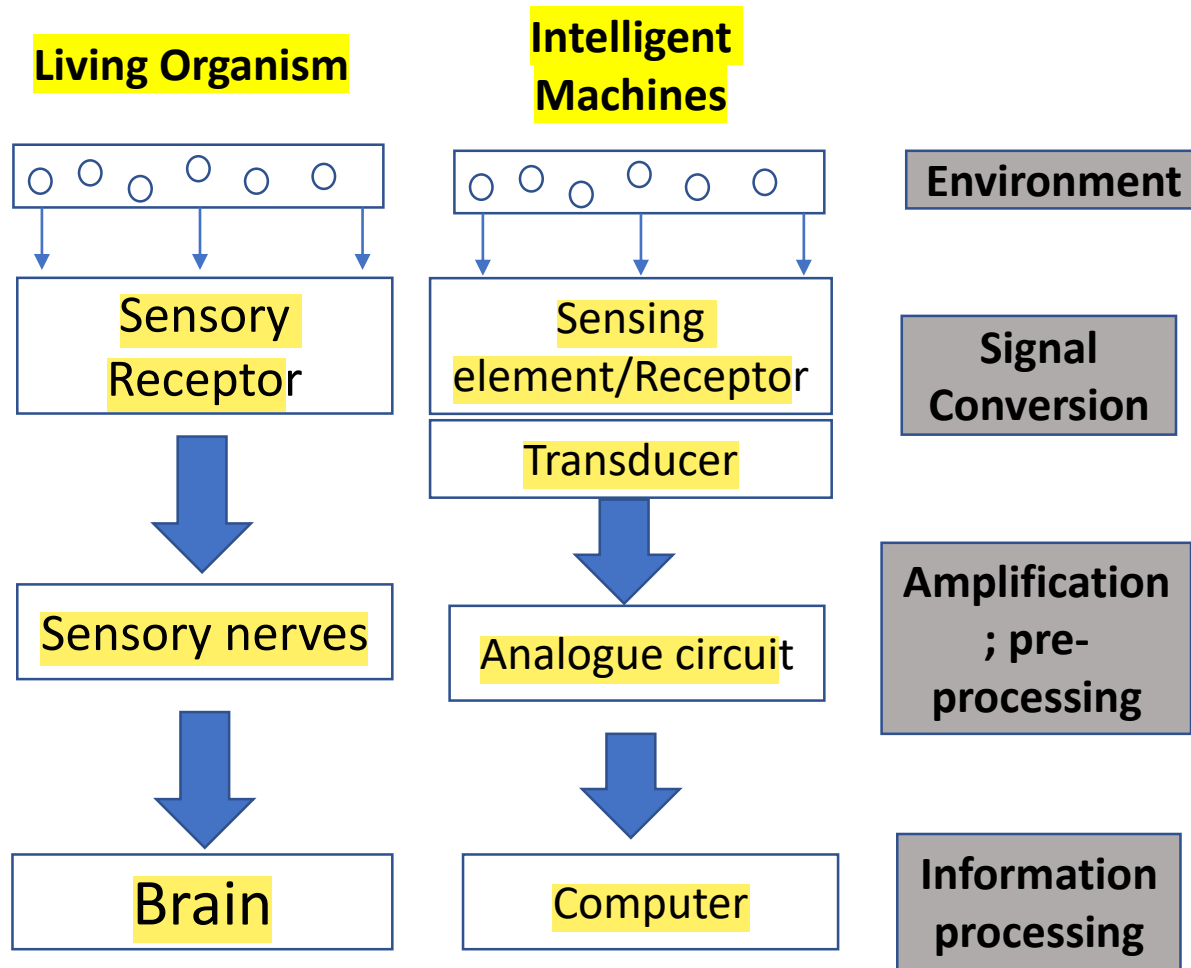
Frequency

Optical



Sensors: Introduction

A **sensor** is a device, module, machine, or subsystem whose purpose is to **detect** events or changes in its environment and send the information to other electronics



Classification of Sensors

Classification of sensors is accomplished in different ways. Prevalent is a classification following the principles of *signal transduction* (IUPAC 1991). The following sensor groups result:

Optical Sensors Absorbance, reflectance, luminescence, fluorescence, refractive index, optothermal effect and light scattering

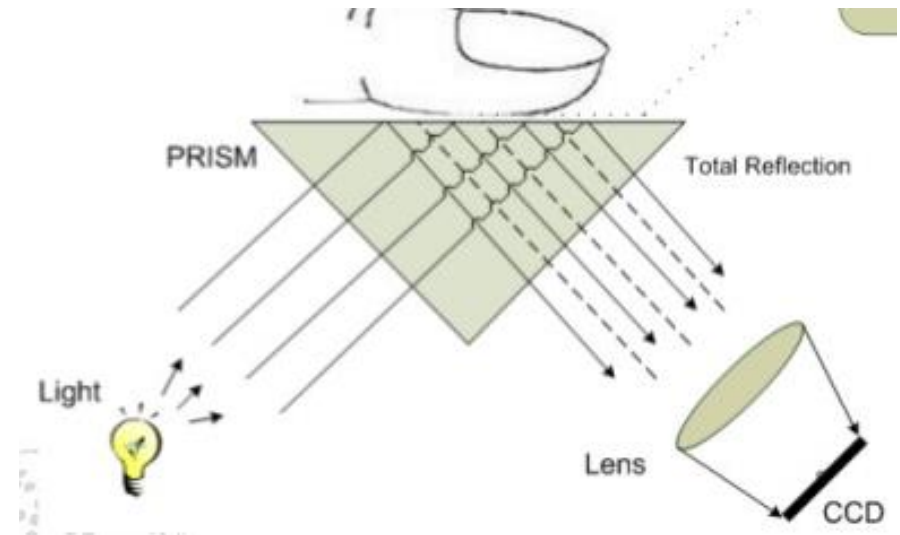
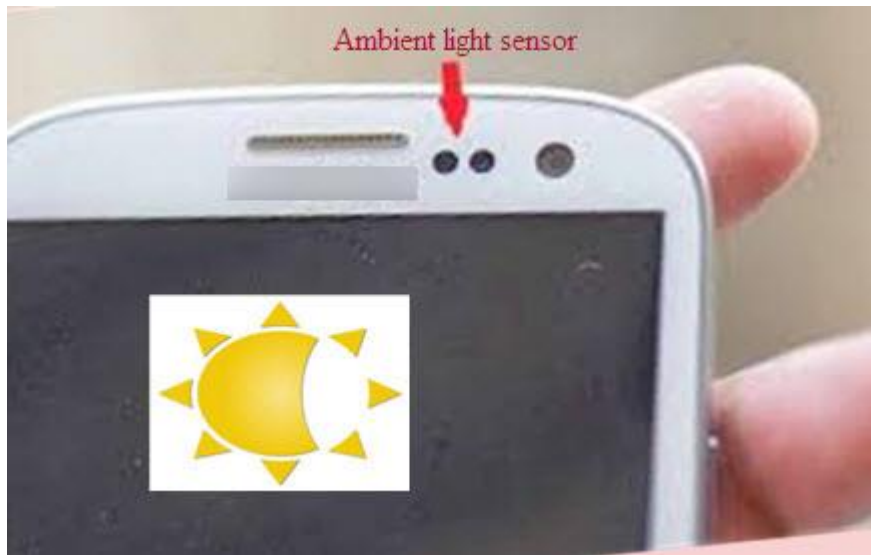
Electrochemical Sensor among them voltammetric and potentiometric devices, chemically sensitized field effect transistor (CHEMFET) and potentiometric solid electrolyte gas sensors

Electrical sensors including those with metal oxide and organic semiconductors as well as electrolytic conductivity sensors.

Thermometric sensors based on the measurement of the heat effect of a specific chemical reaction or adsorption which involves the analyte.

Biosensors are chemical sensors in which the recognition system utilizes a biochemical mechanism.

Optical Sensors



Chemical Sensor

The following **characteristics** of chemical sensors are generally accepted.

Chemical sensors should:

- Transform chemical quantities into electrical signals,
- Respond rapidly,
- Maintain their activity over a long time period,
- Be small,
- Be cheap,
- Be *specific*, i.e. they should respond exclusively to one analyte, or at least be *selective* to a group of analytes.

The above list could be extended with, e.g., the postulation of a *low detection limit*, or a *high sensitivity*. This means that low concentration values should be detected.

Chemical Sensors

Ion Sensors

**Liquid film
ion sensors**

**Solid film
ion sensors**

**ion
selective
FET**

Gas Sensors

**Semiconductor
gas sensor**

**Electrochemical
gas sensor**

**Contact
combustion gas
sensor**

**Polymer gas
sensor**

**Optical gas
sensor**

Humidity sensors

**Resistance
sensor**

**Capacitive
sensor**

Other

Biosensors

**Microbial
sensor**

**Enzyme
sensor**

**Immuno
sensor**

With so many different sensors described in the literature one may ask:

which one is the best?

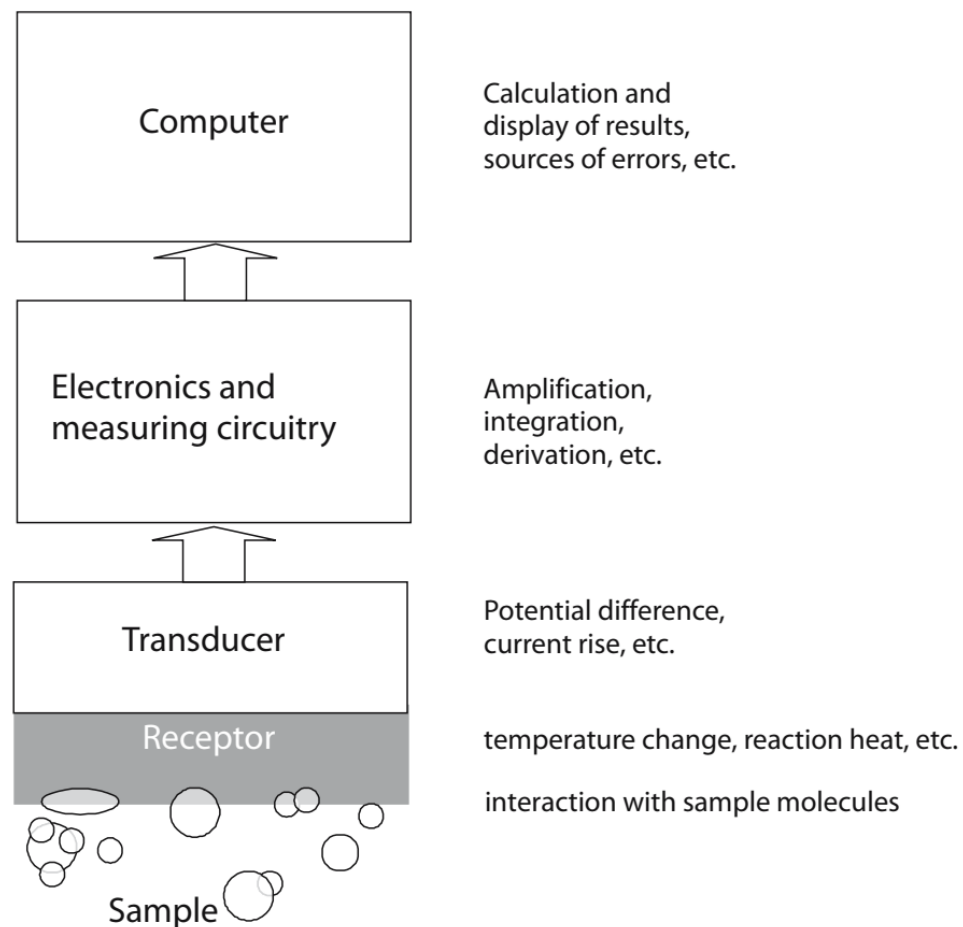
Of course, there is no simple answer, because there are so many *different sensing situations and criteria* that must be considered. Instead, a general precept must serve. Sensor should:

- Be *indirect contact* with the investigated subject
- Transform *non-electric information into electric signals*
- Respond *quickly*
- Operate *continuously* or at least in repeated cycles
- Be *small*
- Be *cheap*
- Be *specific*, i.e. they should respond exclusively to one analyte, or at least be selective to a group of analytes

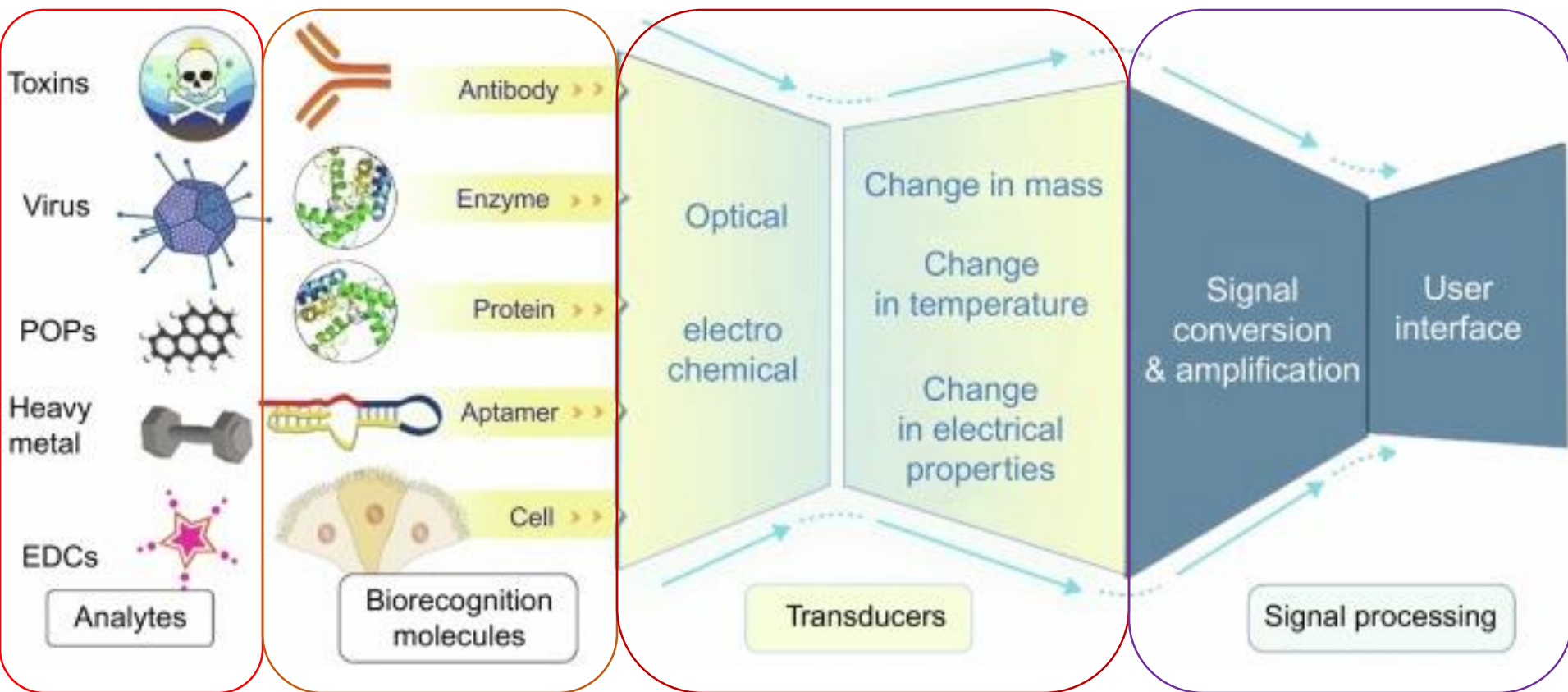
Components of a Chemical Sensor

Chemical sensors usually contain two basic components connected in series: *a chemical (molecular) recognition system (receptor)* and *a physicochemical transducer*.

In the majority of chemical sensors, the receptor interacts with analyte molecules. As a result, its physical properties are changed in such a way that the appending transducer can gain an electrical signal.



Components of a Chemical Sensor



Analytes

Receptor

Transducer

**Signal processing
unit**

Receptor

- The function of the receptor is fulfilled in many cases by a thin layer which is able to **interact with the analyte** molecules, catalyze a reaction selectively, or participate in a chemical equilibrium together with the analyte.
- The receptor layer can respond selectively to particular substances or to a group of substances. The term **molecular recognition** is used to describe this behavior.
- Among the interaction processes, the most important for chemical sensors are **adsorption, ion exchange etc.**
- Primarily these phenomena act at the interface between analyte and receptor surface.

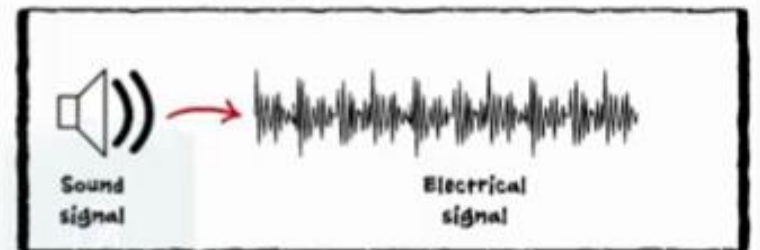
Receptor-Analyte Interaction Equilibria on Surfaces

Receptor layers can respond selectively to particular substances or to a group of substances (**analytes**)

The term **molecular recognition** is used to describe this behavior.

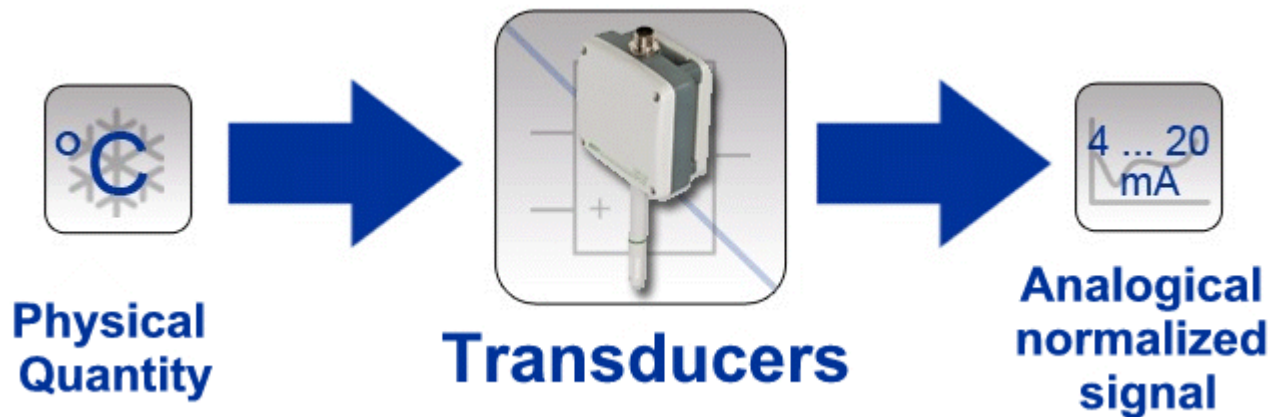
Processes at the receptor-analyte interface can be classified into **interaction equilibria** and **chemical reaction equilibria**

Transducer



Transducer

A device that converts one form of energy into another is called a transducer



a non-electric quantity must be transformed into an electric quantity, voltage, current or resistance.

Transducer

- Nowadays, signals are processed almost exclusively by means of **electrical instrumentation**.
- Accordingly, every sensor should include a transducing function, i.e. **the actual concentration value, a non-electric quantity must be transformed into an electric quantity, voltage, current or resistance.**
- Some of them develop their sensor function only in combination with an additional receptor layer.
- In other types, receptor operation is an inherent function of the transducer.

Semiconductors Surfaces as Receptors & Transducers

In some cases, one and the same physical object acts *as receptor and as transducer*. This is the case e.g. in metallic oxide *semiconductor* gas sensors which change their electrical conductivity in contact with some gases

Example: **Glucose Detector**

- Blood glucose monitors are devices that are used to measure the concentration of glucose within a person's blood.
- Along with the term blood glucose this reading is also often referred to as blood sugar.
- Most basic blood glucose monitors have three separate parts that are required for them to work properly:

a lancer,

a test strip

the meter

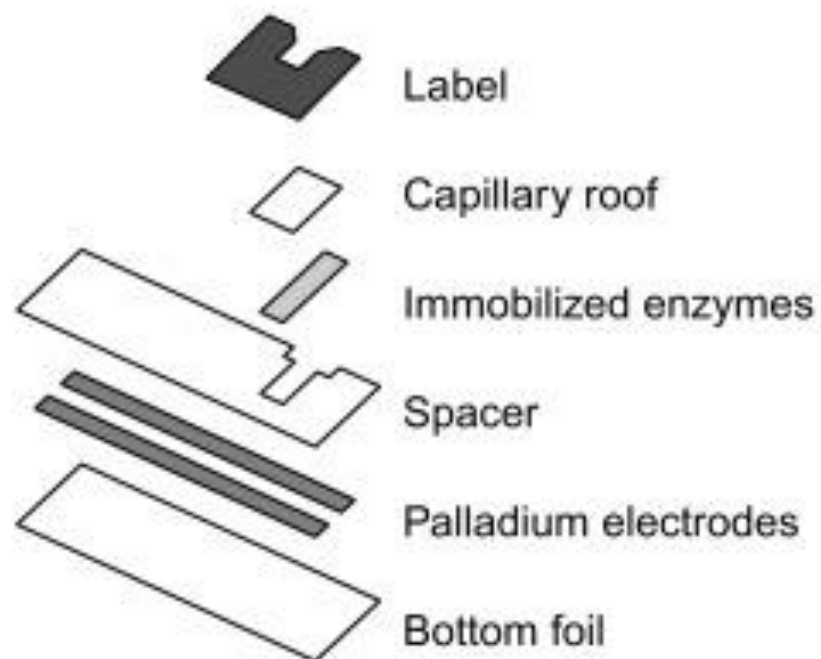


Glucose Detector- working

- Within the test strip the blood is mixed with **glucose oxidase**, which reacts with the **glucose** in the blood sample to create **gluconic acid**.
- Another chemical within the test strip, called **ferricyanide** $[\text{Fe}(\text{CN})_6]^{3-}$, then reacts with the gluconic acid to create **ferrocyanide** $[\text{Fe}(\text{CN})_6]^{4-}$.
- The mediator in turn is reoxidized by reaction at the electrode, which generates an electric current.
- The total charge passing through the electrode is **proportional to the amount of glucose in the blood** that has reacted with the enzyme.
- The electrode within the test strip then runs a current through the blood sample and the ferrocyanide influences this current in such a way that the **concentration of blood glucose** within the sample can be accurately measured within a fair margin of error



(A)



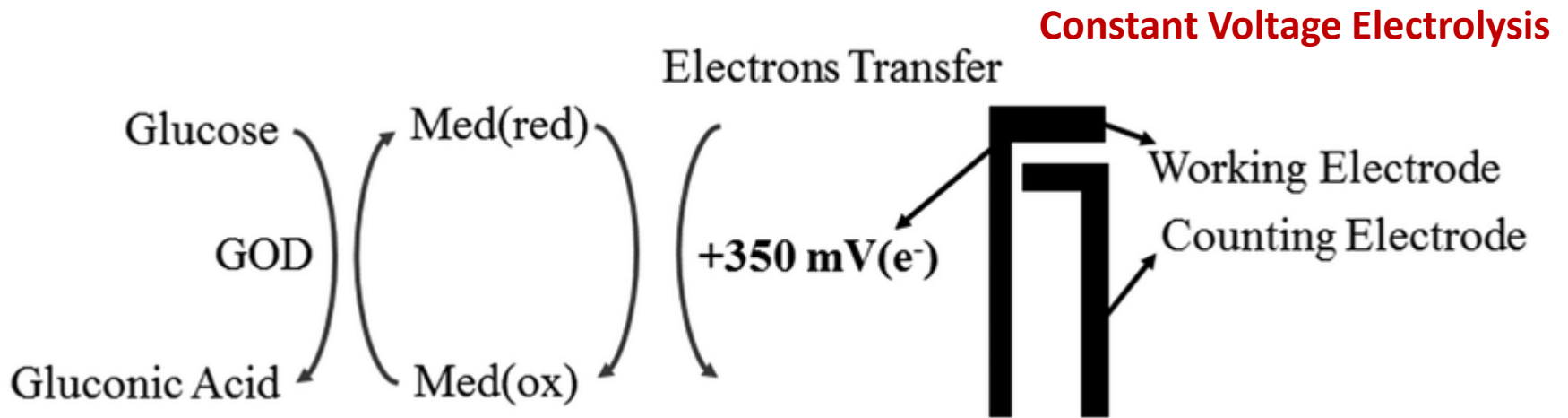
(B)



Working principle of GOD-based glucose electrochemical sensor

Second generation glucose sensors;

O₂ is replaced with potassium ferricyanide as mediator or electron acceptor



GOD = Glucose oxidase

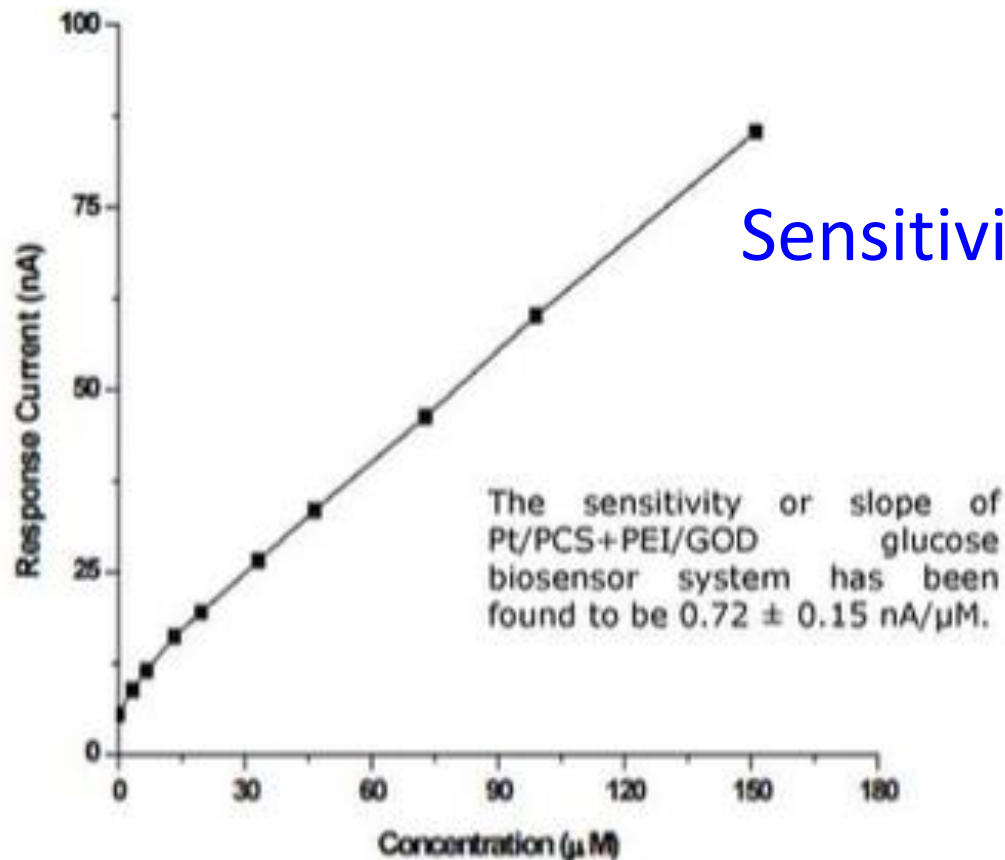
Med = Redox mediator = K₃[Fe(CN)₆]

GOD is **highly selective to glucose**. Since it is an enzyme catalyzed reaction, the response time of the sensor is quick.

Unlike H₂O₂, potassium ferricyanide is **not harming GOD** and its quantity can be altered to suit the application.

Typical Calibration Curve For Glucose Sensor

Response current almost linearly increases with concentration of glucose.



Sensitivity = slope

High sensitivity means, a small change in concentration of analyte produces a large change in transduced signal (e.g. current)

Fig: Calibration curve of the enzyme electrode for glucose

Parameters of Chemical Sensor

- **Sensitivity:** change in the measurement signal per concentration unit of the analyte, i.e. the slope of a calibration graph.
- **Selectivity:** an expression of whether a sensor responds *selectively* to a group of analytes or even *specifically* to a single analyte. Quantitative expressions of selectivity exist for different types of sensors.
- **Detection limit:** the **lowest concentration** value which can be detected by the sensor in question, under definite conditions. Whether or not the analyte can be quantified at the detection limit is not determined. Procedures for evaluation of the detection limit depend on the kind of sensor considered.
- **Dynamic range:** the concentration range between the detection limit and the upper limiting concentration.

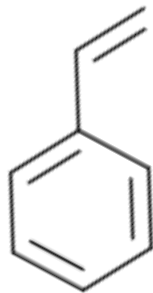
Parameters of Chemical Sensor

- **Resolution:** the **lowest concentration difference** which can be distinguished when the composition is varied continuously. This parameter is important chiefly for detectors in flowing streams.
- **Response time:** the time for a sensor to respond from zero concentration to a step change in concentration. Usually specified as the time to rise to a definite ratio of the final value.
- **Stability:** the ability of the sensor to maintain its performance for a certain period of time. As a measure of stability, drift values are used, e.g. the signal variation for zero concentration.
- **Life cycle:** the length of time over which the sensor will operate. The maximum storage time (*shelf life*) must be distinguished from the maximum *operating life*. The latter can be specified either for continuous operation or for repeated on-off cycles.

Vizag gas leak

LG polymer chemical plant

Styrene monomer



Styrene monomer must be stored between 20–22 °C, above which it will rapidly vaporize

